

# A Certified Lower Bound for Lebesgue’s Universal Cover Problem

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## Abstract

Lebesgue’s universal cover problem asks for a planar set of least possible area that contains a congruent copy of every planar set of diameter at most one. We work in the convex Brass–Sharifi three-test-set framework, where the test sets are a disk, an equilateral triangle, and a regular pentagon of diameter one. For each normalized placement  $v$ , let  $A(v)$  denote the area of the convex hull of these three test sets. We construct a finite certificate proving

$$A(v) \geq 0.83201$$

throughout the admissible normalized domain. The threshold 0.83201 slightly improves the Brass–Sharifi lower bound 0.832 within the same convex three-test-set framework.

The proof is a finite-cover argument. The admissible domain is covered by finitely many parameter domains, and each domain carries a local lower-bound certificate. Most domains are handled by supporting local records. On the witness domains, the local bound is obtained from an inner-witness polygon construction. The witness points lie in the three test sets and determine an ordered polygonal region certified to be simple and positively oriented. Its area is bounded below by interval orientation and shoelace estimates. Since this certified polygonal region lies inside the corresponding convex hull, its area gives a lower bound for the hull area. Combining the local inequalities with the finite cover yields

$$\alpha_{\text{cvx}} \geq 0.83201,$$

where  $\alpha_{\text{cvx}}$  is the infimum of the areas of convex universal covers.

## 1 Introduction

A planar set is a universal cover if it contains a congruent copy of every planar set of diameter at most one. In this paper the covering set is required to be convex. Let

$$\mathcal{U}_{\text{cvx}} = \left\{ K \subset \mathbb{R}^2 : \begin{array}{l} K \text{ is convex, and for every } S \subset \mathbb{R}^2 \text{ with } \text{diam}(S) \leq 1, \\ \text{there is a Euclidean isometry } g \text{ such that } g(S) \subseteq K \end{array} \right\}.$$

Set

$$\alpha_{\text{cvx}} = \inf_{K \in \mathcal{U}_{\text{cvx}}} \text{area}(K).$$

All lower-bound statements below concern this convex quantity. For background on the general universal cover problem, see Baez, Bagdasaryan, and Gibbs [2]; a small convex covering construction is described by Gibbs [3].

Brass and Sharifi introduced the three-test-set framework used here and obtained the lower bound 0.832 in this convex setting [1]. The test sets are a disk, an equilateral triangle, and a regular pentagon of diameter one. Their normalization turns the relative placement problem into a

finite-dimensional problem for the area of the convex hull of these three sets. The present work certifies the threshold

$$\tau = 0.83201$$

within the same normalized convex setting. We regard 0.83201 as an exact decimal threshold.

Although the numerical gain is modest, the finite certificate gives a reproducible verification of the improved bound. The proof is represented by finite records. These include supporting local records for most domains, explicit witness-domain polygon certificates on the witness domains, interval orientation and shoelace lower endpoints, and a final finite-cover aggregation. The accompanying finite record contains the checks used in the proof, and the verification code replays them deterministically.

The structure of the proof is as follows. First, a convex universal cover contains congruent copies of the three test sets, and convexity forces it to contain their convex hull. Second, the Brass–Sharifi normalization represents the hull by a parameter  $v$  in an admissible domain  $\Omega_{\text{adm}}$ . Third, the certificate covers  $\Omega_{\text{adm}}$  by finitely many domains and verifies a local lower bound on each one. Fourth, the finite-cover implication gives  $A(v) \geq \tau$  throughout  $\Omega_{\text{adm}}$ , and taking the infimum over convex universal covers gives  $\alpha_{\text{cvx}} \geq \tau$ .

*Outline.* Section 2 defines the normalized three-test-set problem. Section 3 states the finite certificate model. Section 4 explains the sources of local lower bounds. Section 5 gives the inner-witness polygon construction. Section 6 describes the interval orientation and shoelace estimates. Section 7 proves the certified lower-bound theorem and the convex consequence. Section 8 concludes and records the scope of the statement. Appendices A and B summarize the finite certificate data used by the proof.

## 2 The normalized three-test-set problem

### 2.1 The three test sets

Let  $C$ ,  $T$ , and  $P_5$  be, respectively, fixed reference copies of a disk, an equilateral triangle, and a regular pentagon of diameter one. The reference disk is centered at the origin. The reference triangle and pentagon have fixed reference positions and orientations. The three sets  $C, T, P_5$  are the necessary test sets used in the lower-bound argument: every universal cover contains a congruent copy of each of them. Thus, if  $K \in \mathcal{U}_{\text{cvx}}$ , then  $K$  contains three such copies; since  $K$  is convex, it also contains their convex hull.

### 2.2 Normalized placements

Following the Brass–Sharifi normalization, the disk is fixed at the origin and the orientation of the triangle is fixed. The regular pentagon is taken in a fixed reference orientation; its rotational symmetry permits the pentagon angle to be recorded in a fundamental interval. Let  $R_\rho$  denote counterclockwise rotation by angle  $\rho$  about the origin. Let

$$v = (\rho, x_3, y_3, x_5, y_5), \quad u_3 = (x_3, y_3), \quad u_5 = (x_5, y_5).$$

Here  $u_3$  is the triangle translation,  $\rho$  is the pentagon rotation, and  $u_5$  is the pentagon translation. Set

$$X(v) = C \cup (T + u_3) \cup (R_\rho P_5 + u_5), \quad H(v) = \text{conv } X(v),$$

and set

$$A(v) = \text{area}(H(v)).$$

Here  $\text{conv}$  denotes the closed convex hull. Thus  $A$  denotes the hull-area function, while  $\text{area}$  denotes planar area. Figure 1 illustrates the normalized placement.

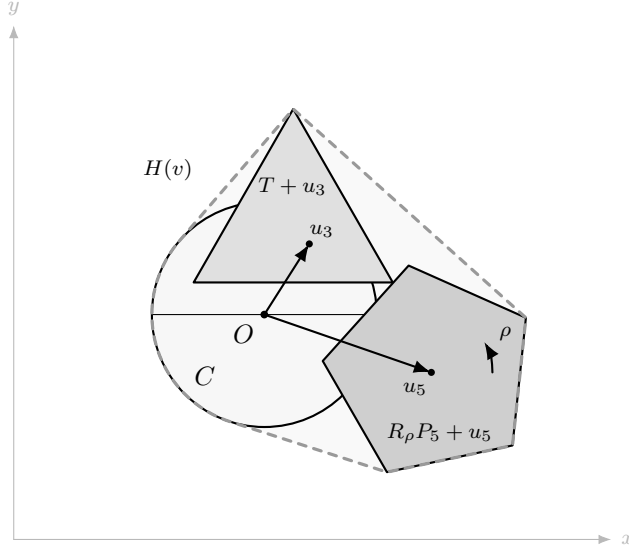


Figure 1: Schematic normalized placement of the three Brass–Sharifi test sets. The disk  $C$  is fixed at the origin  $O$ , the triangle is translated by  $u_3 = (x_3, y_3)$ , and the pentagon is rotated counterclockwise about the origin by  $\rho$  and then translated by  $u_5 = (x_5, y_5)$ . The enclosing curve represents  $H(v) = \text{conv}(C \cup (T + u_3) \cup (R_\rho P_5 + u_5))$ .

### 2.3 The admissible domain and normalization principle

Denote by  $\Omega_{\text{adm}}$  the admissible normalized placement domain in the Brass–Sharifi framework. This is a recorded subset of the five-dimensional parameter space with coordinates  $(\rho, x_3, y_3, x_5, y_5)$ . Its points represent normalized relative placements satisfying the Brass–Sharifi admissibility conditions after the normalization and symmetry reductions used for the three test sets.

The present paper uses the Brass–Sharifi normalization as an input theorem and treats this recorded admissible-domain object as the input domain for the finite certificate. It does not rederive the full symbolic reduction from the unreduced placement space. The finite cover of  $\Omega_{\text{adm}}$  is introduced in Section 3.

We use the following normalization principle from the Brass–Sharifi framework: for every  $K \in \mathcal{U}_{\text{cvx}}$ , there exists  $v \in \Omega_{\text{adm}}$  such that

$$H(v) \subseteq K.$$

Consequently, a uniform lower bound for  $A(v)$  on  $\Omega_{\text{adm}}$  gives a lower bound for  $\text{area}(K)$  for every convex universal cover  $K$ . This is the only point at which the Brass–Sharifi normalization enters the final universal-cover consequence. The certificate proves

$$A(v) \geq \tau \quad (v \in \Omega_{\text{adm}}),$$

and the convex universal-cover consequence is proved in Section 7.

## 3 The finite certificate model

**Definition 1** (Finite certificate). *A finite certificate for the threshold  $\tau$  consists of the following finite data.*

- (i) A finite family  $\mathcal{F}$  of parameter domains in the normalized placement space.
- (ii) A cover assertion  $\Omega_{\text{adm}} \subseteq \bigcup_{B \in \mathcal{F}} B$ .
- (iii) For each  $B \in \mathcal{F}$ , a local value  $L_B$  satisfying  $A(v) \geq L_B \geq \tau$  on  $B$ .
- (iv) Individual evidence records certifying the local assertions.
- (v) Outward-rounded interval enclosures for the determinant and area estimates used by the certificate.
- (vi) A final aggregation check connecting the local assertions to the convex lower-bound statement.

### 3.1 Certificate cover

Let  $\mathcal{F}$  be the finite cover supplied by the certificate. It is a finite family of parameter domains in the five-dimensional normalized placement space, and each domain  $B \in \mathcal{F}$  is represented by interval bounds on the placement parameters. Its mathematical role is the inclusion

$$\Omega_{\text{adm}} \subseteq \bigcup_{B \in \mathcal{F}} B.$$

The symbol  $\mathcal{F}$  always denotes the cover family, while  $B$  denotes one of its domains.

### 3.2 Local lower-bound assertions

For each  $B \in \mathcal{F}$ , the local assertion has the form

$$A(v) \geq L_B \quad (v \in B),$$

where the certified lower endpoint also satisfies

$$L_B \geq \tau.$$

Section 4 separates the cover according to how the value  $L_B$  is certified; both subfamilies enter the final proof through the same inequality.

### 3.3 Evidence records and interval enclosures

Each local assertion is accompanied by finite evidence. The record families are summarized in Table 1.

Table 1: Certificate record families and their local lower-bound interfaces.

| Record family             | Role in the proof                                           | Local interface                        |
|---------------------------|-------------------------------------------------------------|----------------------------------------|
| directed interval records | Direct interval lower bounds on supporting domains.         | $A(v) \geq L_B \geq \tau$              |
| tensor records            | Auxiliary finite local-bound records on supporting domains. | $A(v) \geq L_B \geq \tau$              |
| bridge records            | Supporting records for residual domains.                    | $A(v) \geq L_B \geq \tau$              |
| witness records           | Ordered witness polygons and interval area lower bounds.    | $A(v) \geq L_B^{\text{wit}} \geq \tau$ |

All supporting record families enter the proof only through certified values  $L_B \geq \tau$ . The use of individual evidence records ties each local assertion in the cover to a checked record, rather than only to an aggregate count.

The numerical records use outward-rounded interval arithmetic. Each coordinate, determinant, and shoelace expression is evaluated in an interval enclosure rounded outward, so a recorded lower endpoint is a rigorous lower bound under the stated interval model. This is the standard principle of interval arithmetic; see Moore [4].

The sizes of the finite cover and the supporting evidence records are summarized in Table A.1. The finite certificate conditions used in the final aggregation are listed in Table A.2. These tables summarize checked certificate data; the proof uses the corresponding local inequalities, not the table entries as independent assumptions. The aggregation step is isolated as Proposition 1. We next separate the local assertions according to the mechanism that supplies  $L_B$ .

## 4 Sources of local lower bounds

Let  $\mathcal{F}_{\text{wit}} \subseteq \mathcal{F}$  be the subfamily of witness domains, and set

$$\mathcal{F}_{\text{sup}} = \mathcal{F} \setminus \mathcal{F}_{\text{wit}}.$$

For  $B \in \mathcal{F}_{\text{sup}}$ , the value  $L_B$  is supplied by supporting interval records and the associated individual evidence records. For  $B \in \mathcal{F}_{\text{wit}}$ , the value  $L_B$  is realized by a witness lower bound  $L_B^{\text{wit}}$ . The distinction between  $\mathcal{F}_{\text{sup}}$  and  $\mathcal{F}_{\text{wit}}$  concerns only the method used to certify the local lower bound; both families enter the final proof through the same inequality  $A(v) \geq L_B \geq \tau$ .

### 4.1 Supporting domains

On the supporting domains  $B \in \mathcal{F}_{\text{sup}}$ , each supporting record certifies an interval lower bound for  $A(v)$  on its domain and checks that the certified lower endpoint is at least the same threshold  $\tau$  used in the theorem. The final proof uses the local inequalities attached to these records, not their aggregate counts. A supporting local record enters the finite cover only after the inequalities  $A(v) \geq L_B$  on  $B$  and  $L_B \geq \tau$  have been checked.

### 4.2 Witness domains

On the witness domains  $B \in \mathcal{F}_{\text{wit}}$ , the local value is obtained from an inner-witness polygon. In this case the local lower bound is the witness lower bound,

$$L_B = L_B^{\text{wit}}, \quad A(v) \geq L_B^{\text{wit}} \quad (v \in B),$$

and the interval estimates of Section 6 establish  $L_B^{\text{wit}} \geq \tau$ . The witness-domain counts and interval lower-bound minima are summarized in Tables B.1 and B.2.

## 5 The inner-witness polygon construction

We now spell out the geometric construction used on the witness domains.

### 5.1 Witness point families

Fix  $B \in \mathcal{F}_{\text{wit}}$ . A witness family on  $B$  is an ordered list of point functions

$$q_1(v), q_2(v), \dots, q_{m_B}(v).$$

We denote the underlying set of these witness points by

$$Q_B(v) = \{q_1(v), \dots, q_{m_B}(v)\}.$$

The containment condition is

$$q_i(v) \in X(v) \quad (1 \leq i \leq m_B, v \in B).$$

Each witness point has recorded provenance: it is a translated vertex of  $T$ , a rotated-translated vertex of  $P_5$ , or a certified point of  $C$ , meaning a point whose membership in the diameter-one disk is verified by outward-rounded interval enclosures. The triangle witness coordinates are affine in the translation variables. The pentagon witness coordinates are trigonometric functions of the rotation angle  $\rho$  and affine functions of the translation  $u_5$ . In all cases, the coordinate enclosures on  $B$  are evaluated by outward-rounded interval arithmetic.

## 5.2 The ordered witness polygon

The certificate also stores a prescribed cyclic order of the witness points. Let  $\mathcal{P}_B(v)$  denote the compact polygonal region bounded by the ordered vertices

$$q_1(v), q_2(v), \dots, q_{m_B}(v).$$

The certificate includes a convex-order audit for this prescribed order. The determinant estimates of Section 6 are not used alone to infer global simplicity from consecutive turns; rather, they certify the stability of the prescribed order on the whole parameter domain. Together, the convex-order audit and the determinant estimates certify that the ordered polygonal region is simple and positively oriented throughout  $B$ .

All vertices of  $\mathcal{P}_B(v)$  lie in  $X(v) \subseteq H(v)$ . Since  $H(v)$  is convex and  $\mathcal{P}_B(v)$  is the polygonal region spanned by these ordered vertices,

$$\mathcal{P}_B(v) \subseteq H(v).$$

## 5.3 The inner-witness lower-bound lemma

**Lemma 1** (Inner-witness lower bound). *Suppose that the ordered witness polygonal region  $\mathcal{P}_B(v)$  is contained in  $H(v)$  for every  $v \in B$ , and that*

$$\text{area}(\mathcal{P}_B(v)) \geq L_B^{\text{wit}} \quad (v \in B).$$

Then

$$A(v) \geq L_B^{\text{wit}} \quad (v \in B).$$

*Proof.* For  $v \in B$ , the containment  $\mathcal{P}_B(v) \subseteq H(v)$  and area monotonicity give

$$A(v) = \text{area}(H(v)) \geq \text{area}(\mathcal{P}_B(v)) \geq L_B^{\text{wit}}.$$

□

## 6 Interval orientation and shoelace estimates

### 6.1 Coordinate enclosures

Write

$$q_i(v) = (x_i(v), y_i(v)), \quad 1 \leq i \leq m_B.$$

The certificate evaluates interval enclosures for these coordinate functions on each witness domain  $B$ . The enclosures are rounded outward at every operation. Consequently, all determinant and shoelace lower endpoints used below are conservative.

### 6.2 Cyclic order and determinant lower bounds

For consecutive triples in the prescribed cyclic order stored in the certificate data, define

$$\Delta_{B,i}(v) = \det(q_{i+1}(v) - q_i(v), q_{i+2}(v) - q_{i+1}(v)),$$

with indices taken cyclically. The interval evaluation supplies lower endpoints  $\underline{\Delta}_{B,i}$  such that

$$\underline{\Delta}_{B,i} \leq \Delta_{B,i}(v) \quad (v \in B).$$

The convex-order audit fixes the cyclic order. For the estimates required by that prescribed order, the recorded condition is

$$\underline{\Delta}_{B,i} > 0.$$

These determinant conditions certify that the prescribed order remains positively oriented on  $B$ . They are used together with the convex-order audit, not as an independent global-simplicity theorem. No shoelace lower bound is accepted unless the convex-order audit and the corresponding determinant lower bounds have both been certified.

### 6.3 Shoelace lower endpoints

For the certified order, set

$$S_B(v) = \sum_{i=1}^{m_B} (x_i(v)y_{i+1}(v) - x_{i+1}(v)y_i(v)), \quad q_{m_B+1} = q_1$$

by cyclic convention. Since  $\mathcal{P}_B(v)$  is certified to be a simple positively oriented polygonal region, the shoelace formula gives

$$\text{area}(\mathcal{P}_B(v)) = \frac{1}{2} S_B(v).$$

The interval computation supplies a lower endpoint  $\underline{S}_B$  satisfying

$$\underline{S}_B \leq S_B(v) \quad (v \in B).$$

Thus

$$\text{area}(\mathcal{P}_B(v)) \geq \frac{1}{2} \underline{S}_B.$$

If  $\frac{1}{2} \underline{S}_B \geq \tau$ , Lemma 1 gives  $A(v) \geq \tau$  on  $B$ . Thus every witness domain carries a verified local inequality  $A(v) \geq \tau$ , placing it in the local framework of Section 3.2.

## 7 The certified lower-bound theorem

### 7.1 Finite-cover proposition

**Proposition 1** (Finite-cover implication). *Let  $\mathcal{F}$  be a finite family of parameter domains such that*

$$\Omega_{\text{adm}} \subseteq \bigcup_{B \in \mathcal{F}} B.$$

*Assume that for every  $B \in \mathcal{F}$  there is a certified number  $L_B$  satisfying*

$$A(v) \geq L_B \geq \tau \quad (v \in B).$$

*Then  $A(v) \geq \tau$  for every  $v \in \Omega_{\text{adm}}$ .*

*Proof.* Let  $v \in \Omega_{\text{adm}}$ . By the finite-cover relation,  $v \in B$  for at least one  $B \in \mathcal{F}$ . The local certificate on that domain gives

$$A(v) \geq L_B \geq \tau.$$

Since  $v$  was arbitrary, the inequality holds on  $\Omega_{\text{adm}}$ .  $\square$

### 7.2 Verification of the bundled certificate

**Proposition 2** (Verification of the bundled certificate). *The certificate archive accompanying this paper satisfies the finite certificate conditions OB-A–OB-F listed in Table A.2 for the threshold  $\tau$ .*

*Proof.* The accompanying finite record is organized into four components summarized in Tables A.1–B.2:

- per-record evidence, which ties supporting local records to individual evidence;
- construction records, which provide the supporting construction and rounding checks;
- witness records, which provide point containment, convex-order audit, determinant lower bounds, and shoelace lower endpoints on the witness domains;
- final aggregation records, which connect the cover-level proof obligations to the stated convex claim boundary.

The hash manifest fixes the record contents used for these checks. The bundled records satisfy all listed conditions, so OB-A–OB-F hold for  $\tau$ .  $\square$

### 7.3 Certified lower-bound theorem

**Theorem 1** (Certified lower-bound theorem). *For the bundled finite certificate, one has*

$$A(v) \geq \tau \quad (v \in \Omega_{\text{adm}}).$$

*Proof.* By Proposition 2, the bundled certificate satisfies the cover condition and the local lower-bound conditions. For  $B \in \mathcal{F}_{\text{sup}}$ , the supporting records give  $A(v) \geq L_B \geq \tau$  on  $B$ . For  $B \in \mathcal{F}_{\text{wit}}$ , Lemma 1 and the interval orientation and shoelace conditions give  $A(v) \geq L_B^{\text{wit}} \geq \tau$  on  $B$ . Thus every domain in  $\mathcal{F}$  satisfies the local hypothesis of Proposition 1, and the theorem follows.  $\square$



## 7.4 Convex universal-cover consequence

**Corollary 1** (Convex universal-cover consequence). *The bundled finite certificate gives*

$$\alpha_{\text{cvx}} \geq \tau = 0.83201.$$

*Proof.* Let  $K \in \mathcal{U}_{\text{cvx}}$ . By the normalization principle of Section 2.3, choose  $v \in \Omega_{\text{adm}}$  such that  $H(v) \subseteq K$ . Then

$$\text{area}(K) \geq \text{area}(H(v)) = A(v).$$

Theorem 1 gives  $A(v) \geq \tau$ . Hence  $\text{area}(K) \geq \tau$ . Taking the infimum over  $K \in \mathcal{U}_{\text{cvx}}$  yields the desired inequality.  $\square$

## 8 Conclusion

The finite certificate proves  $A(v) \geq \tau$  on the admissible normalized three-test-set domain. The threshold  $\tau = 0.83201$  is obtained in the same convex three-test-set framework as the Brass–Sharifi threshold 0.832.

The proof proceeds by finite covering. The admissible domain is covered by finitely many parameter domains, each equipped with a local lower-bound certificate. On the witness domains, the certificate uses ordered inner-witness polygonal regions whose areas are bounded below by interval orientation and shoelace estimates. The remaining domains enter through supporting local records. The finite-cover proposition then yields the convex universal-cover consequence  $\alpha_{\text{cvx}} \geq 0.83201$ .

The statement proved here is the convex lower-bound consequence obtained within the stated Brass–Sharifi three-test-set certificate model. It is not a statement about the unrestricted nonconvex problem, nor is it a proof-assistant formalization.

**Certificate archive and verification code.** The certificate records and verification code accompanying this paper are available from the project repository.<sup>1</sup>

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## References

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<sup>1</sup><https://github.com/Sheldon-Anderson/lebesgue-universal-cover-new-lower-bound-certificate>

## A Finite certificate data

Table A.1: Finite certificate components.

| Component                 | Quantity | Role                             |
|---------------------------|----------|----------------------------------|
| parameter domains         | 356,816  | cover of $\Omega_{\text{adm}}$   |
| directed interval records | 41,261   | supporting local bounds          |
| tensor records            | 8,751    | auxiliary local bounds           |
| bridge records            | 282      | residual supporting local bounds |
| witness domains           | 16       | witness-domain local bounds      |

Table A.2: Finite certificate conditions.

| Group | Mathematical assertion                                        | Certificate component /<br>verification record  |
|-------|---------------------------------------------------------------|-------------------------------------------------|
| OB-A  | $\Omega_{\text{adm}} \subseteq \bigcup_{B \in \mathcal{F}} B$ | finite-cover records                            |
| OB-B  | supporting domains satisfy local lower bounds                 | per-record evidence and<br>construction audit   |
| OB-C  | witness domains satisfy local lower bounds                    | witness construction                            |
| OB-D  | every used local record has individual evidence               | per-record evidence                             |
| OB-E  | interval endpoints clear $\tau$                               | construction and witness<br>interval records    |
| OB-F  | final aggregation matches the convex claim                    | final aggregation and<br>claim-boundary records |

## B Witness-domain interval summaries

Table B.1: Witness-domain counts.

| Quantity                       | Value         |
|--------------------------------|---------------|
| witness domains                | 16            |
| accepted terminal subdomains   | 140           |
| unresolved terminal subdomains | 0             |
| witness point incidences       | 2,112         |
| point-containment certificates | 2,112 / 2,112 |

Table B.2: Interval lower-bound summaries on the witness domains.

| Quantity                                       | Value                   |
|------------------------------------------------|-------------------------|
| minimum witness-domain area bound              | 0.8642876791            |
| minimum excess over $\tau$                     | 0.0322776791            |
| minimum orientation determinant lower endpoint | $3.6637 \times 10^{-5}$ |

The minima in Table B.2 are taken only over the witness domains. They are not asserted to be the global infimum of  $A(v)$  over  $\Omega_{\text{adm}}$ ; the theorem uses only that every accepted local record clears the threshold  $\tau$ .